

Achal Sharma

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PROFILE

Passionate Computer Engineering student at **Trinity College Dublin** with diverse experience including professional web development, machine learning research at NIT, and Electronic Arts software engineering simulation, skilled in Python, C++ and full-stack technologies.

SKILLS

Languages: C, C++, Python, ARM Assembly, JavaScript (ES6+), HTML5, CSS3, SQL

Frameworks & Libraries: Flask, React (Basic), RESTful APIs, Socket Programming

Tools & Platforms: Git, GitHub Actions, VS Code, Raspberry Pi, Docker (Familiar), Arduino IDE, MATLAB, MySQL, PostgreSQL

Development Practices: Object-Oriented Programming (OOP), Test-Driven Development (TDD), Agile Methodologies, CI/CD,

Debugging, Technical Documentation, Distributed Systems Concepts

Soft Skills: Excellent communicator, documentation & presentation, ownership mindset, team-oriented, adaptability, fast learner

WORK EXPERIENCE

Web Developer Intern | TechMarbles Web Solutions Pvt. Ltd. | Remote

June 2025 – August 2025

- Developed and maintained 4+ dynamic, responsive web applications using modern JavaScript, HTML5, and CSS3, improving page load times by 15% through front-end optimization techniques.
- Engineered robust back-end solutions and integrated RESTful APIs that enhanced application functionality and reduced data processing time by 25%.
- Collaborated in a professional Agile development environment, utilizing Git for version control across 15+ repositories and participating in 20+ code reviews to maintain code quality standards.

Research Intern | National Institute of Technology (NIT) Manipur | Imphal, India

October 2023 – December 2023

- Conducted intensive research on medical image analysis, processing 500+ MRI scans to develop a brain tumour detection system with 91.5% classification accuracy.
- Engineered a complete machine learning pipeline implementing image preprocessing, feature extraction, and K-Nearest Neighbours algorithm, reducing false positives by 30% through cross-validation optimization.
- Collaborated directly with Dr. Khelehandra Thongam, presenting research findings and contributing to academic documentation for potential publication.
- Reduced manual diagnosis time by 60% through automated classification, demonstrating the practical application of ML in healthcare diagnostics.

Sales Associate | Classdale Limited, Dundrum Mall, Dublin |

May 2025 – Aug 2025

- Developed strong interpersonal and communication skills assisting diverse clientele and managing store operations.

PROJECTS

Full-Stack e-Attendance System | Python, Flask, MySQL, HTML, CSS

- Designed and developed a secure, full-stack web application to automate attendance tracking.
- Implemented a back-end with Python and Flask to handle QR code generation/validation and manage user sessions.
- Built a responsive front-end and integrated a SQL database for persistent data storage, streamlining the check-in/out process.

Distributed TCP Consensus Protocol Simulator | Python, Socket Programming

- Engineered a simulation of a distributed consensus algorithm to handle node communication and data consistency.
- Implemented custom socket programming in Python to manage network interactions, including retry mechanisms and timeout handling for unreliable network conditions.

Embedded Morse Code Game | C, ARM Assembly, Raspberry Pi Pico

- Developed an interactive game firmware from the ground up, showcasing proficiency in low-level programming.
- Designed an interrupt-driven system to handle real-time input from buttons and control outputs to LEDs and a buzzer, demonstrating mastery of hardware-software interaction.

Machine Learning Pipeline for Medical Imaging | Python, Scikit-learn, NumPy, Pandas

- Architected a complete ML pipeline for image classification, including data preprocessing, feature extraction, and model training.
- Implemented a K-Nearest Neighbours classifier from scratch, optimizing its performance through cross-validation and hyperparameter tuning.

CERTIFICATIONS

Software Engineering Job Simulation–Electronic Arts

Sept 2025

Brain Tumour Detection using KNN –NIT Manipur

June 2025

EDUCATION

Trinity College Dublin, Ireland

Expected Graduation in May 2026

Bachelor's degree in computer engineering

Expected first Class Honors, First Division

Relevant Coursework: Probability & Statistics, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Artificial Intelligence, Machine Learning, Deep Learning, Database Systems, Software Design Analysis, Operating Systems, Security & Privacy, Management, Knowledge Engineering, Computer Networks, Project Management, Accounting, Professional Communication